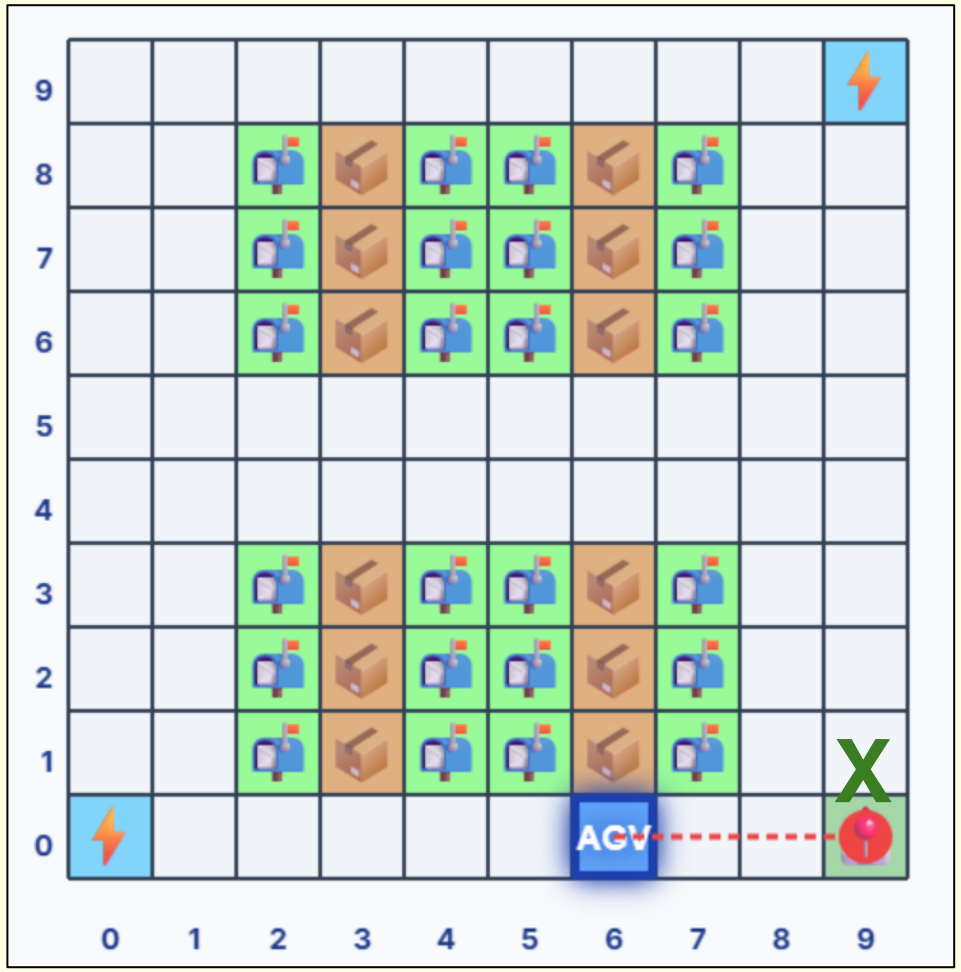
Task Planning for AGV Picking System in Smart Warehousing

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Traffic/Markov/Extremum/LP/

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A. Problem Overview



10-by-10 warehouse mesh:

- Charging points: **(0,0), (9,9)**
- Depot: **(9,0)**
- Picking Zone: **Green Mailbox**

Travelling AGV:

- Random **picking point** → **A** → **X** → **A** → **B** → **X** → **B** →
- Task: → **pick-up** → **deliver** → **replenish** →
- Constraint A: **recharge** needed
- Constraint B: subject to **congestion**

A. Problem Setting (1)

Warehouse mesh:

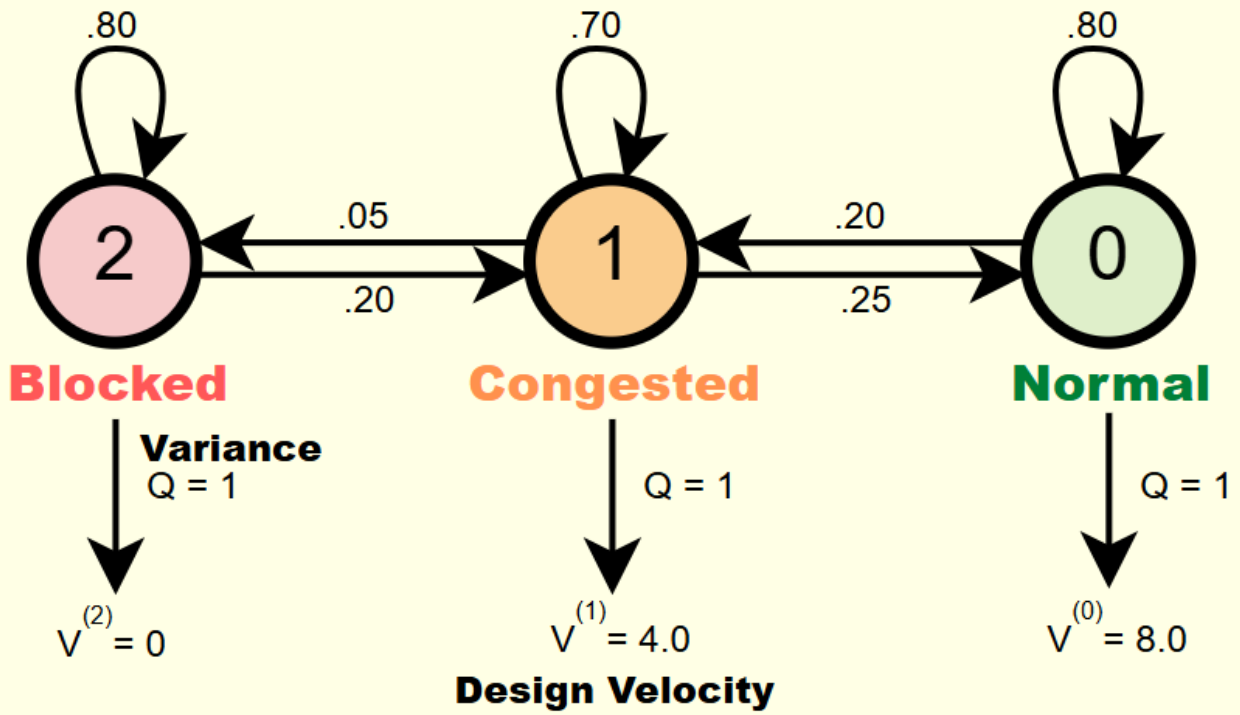
- Charging points **(0,0), (9,9)**: Recharge battery to the full capacity
- Depot **(9,0)** and **Picking Zone**: At each stage, there are three subtasks with a randomly chosen pickup point, which are defined by **pick-up** → **deliver** → **replenish**, then proceed to the next loop
- **Containers**: The area which should be the inaccessible obstacles

Automated Guided Vehicle:

- Traveling between task points and the charging stations to complete as many tasks as possible with the **power** constraint
- Its movement efficiency largely depends on **traffic conditions**

A. Setup – HMM Traffic Mode

Hidden Markov Model (HMM)



$$\bar{\pi} = [.5 \quad .4 \quad .1]$$

Steady-State Vector $\bar{\pi}$

$$\mathbf{M} = \begin{bmatrix} .80 & .20 & .00 \\ .25 & .70 & .05 \\ .00 & .20 & .80 \end{bmatrix}$$

Transfer Matrix M

Emission Relationship

$$v'_k \sim N\left(0.25 v_{k-1} + \mathbf{0.75} v^{(s_k)}, 1\right)$$

$$\underbrace{\begin{pmatrix} \mathbf{25\%}\ prev\ V \\ + \\ \mathbf{75\%}\ design\ V \end{pmatrix}}_{\text{Linear Combination}} + \underbrace{\omega}_{\text{Deviation}}$$

A. Setup – Moving Success Rate

Challenge: How to handle position transfer based on mesh?

Solution: **Discretization!**

$$v_t = v'_t + c_t$$

Controllable Variable

$$(v^{(2)} = 0) \leq v_t \leq (v^{(0)} = 8)$$

$$\begin{cases} P(\Delta x_t = 1) = v_t/8 \\ P(\Delta x_t = 0) = 1 - v_t/8 \end{cases}$$

Bernoulli Trial

A. Problem Setting (2)

Traffic Condition & Movement:

- The traffic condition at each time step is a Markov process, and related conditions are given with a transfer matrix $\boldsymbol{M}$
- The movement between grid is a linear probability function in v
- Under each traffic condition, there is a suggested *design speed* in default. It is the anchor of natural velocity update and affects the **power** consumption as stated in next page.
- The natural velocity update constitutes of 50% velocity at the end of last step, 50% design speed referring to the traffic condition at the new step, and a variance of 1 at all time.

A. Setup – Power Cost & Limit

1. $\Delta P_t = -1 - 0.2\left(v_t - v^{(s_t)}\right)^2 \rightarrow$ Quadratic, at least 1

2. $max(P) = 100 \rightarrow$ Recharging: $P_t = 100$

3. $\widehat{P}_t = \begin{cases} \mathbf{0}: \text{High} \ (40 < P_t \leq 100) \\ \mathbf{1}: \text{Medium} \ (20 < P_t \leq 40) \\ \mathbf{2}: \text{Low} \ (0 < P_t \leq 20) \end{cases} \rightarrow$ Discretization

1st Total **task** completion ↑

2nd Total **energy** consumption ↓

$maximize \ \frac{per\ task}{energy}$!!

A. Problem Setting (3)

Power Consumption:

- At least a fixed cost of 1 and an extra overhead, which is quadratically proportional with the value of velocity deviated from the *design speed*
- The battery need charging at charging stations at **(0, 0)** and **(9, 9)**
- The vehicle will merely get the simplified power level, such as high (40 ~ 100%), medium (20 ~ 40%), and low (0 ~ 20%), to make its moving decision at every step

Anchoring Probability

Find **traffic mode**

Velocity Control

Max. **displacement**
Min. **cost**

Moving Strategy

Recharge or **work**?

B. Probability Overview

Markov & Conditional Probability:

- Based on the nature of state transferring, the solution takes the likelihood of all transfer modes to infer the overall belief of current traffic condition

Post-hoc Regression:

- The comparison of statistical result and belief under the theorical model draws the conclusion that it have high accuracy in assessing the traffic conditions

B. Probability – Likelihood

Probabilistic Ensemble

$$\boldsymbol{Y_k^{(0)}}|X_k:\ \Delta v_k'(\boldsymbol{s=0}|s=i)\sim N(-0.75v_{k-1}+0.75v^{(\boldsymbol{0})})$$

$$\boldsymbol{Y_k^{(1)}}|X_k:\ \Delta v_k'(\boldsymbol{s=1}|s=i)\sim N(-0.75v_{k-1}+0.75v^{(\boldsymbol{1})})$$

$$\boldsymbol{Y_k^{(2)}}|X_k:\ \Delta v_k'(\boldsymbol{s=2}|s=i)\sim N(-0.75v_{k-1}+0.75v^{(\boldsymbol{2})})$$

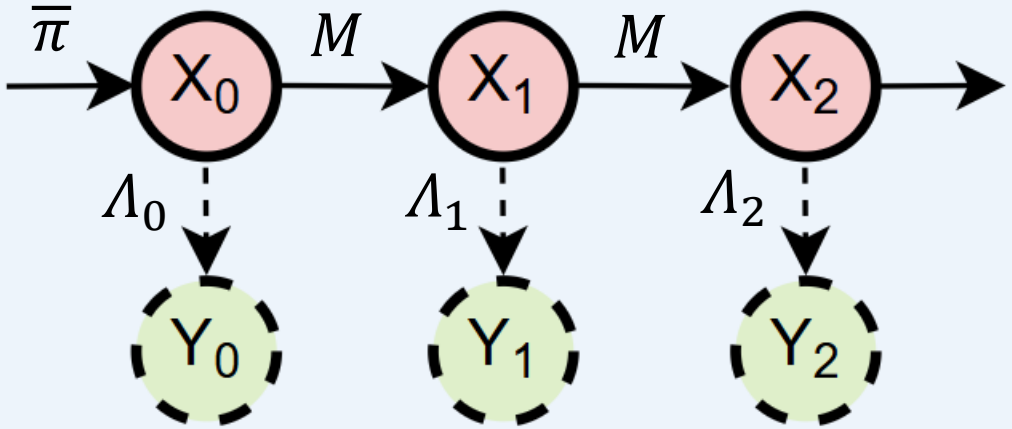
$$\boldsymbol{\Lambda_t}=\begin{bmatrix}\Delta v_t'(\boldsymbol{0}|0) & \Delta v_t'(\boldsymbol{1}|0) & 0\\ \Delta v_t'(\boldsymbol{0}|1) & \Delta v_t'(\boldsymbol{1}|1) & \Delta v_t'(\boldsymbol{2}|1)\\ 0 & \Delta v_t'(\boldsymbol{1}|2) & \Delta v_t'(\boldsymbol{2}|2)\end{bmatrix}$$

Likelihood Matrix $\boldsymbol{\Lambda_t}$

$$\mathbf{M}=\begin{bmatrix}\mathrm{P}(\boldsymbol{0}|0) & \mathrm{P}(\boldsymbol{1}|0) & 0\\ \mathrm{P}(\boldsymbol{0}|1) & \mathrm{P}(\boldsymbol{1}|1) & \mathrm{P}(\boldsymbol{2}|1)\\ 0 & \mathrm{P}(\boldsymbol{1}|2) & \mathrm{P}(\boldsymbol{2}|2)\end{bmatrix}$$

Transfer Matrix $\boldsymbol{M}$

B. Probability – Decoding



(**variable** emission probability)

Hidden Markov Model

Forward Algorithm

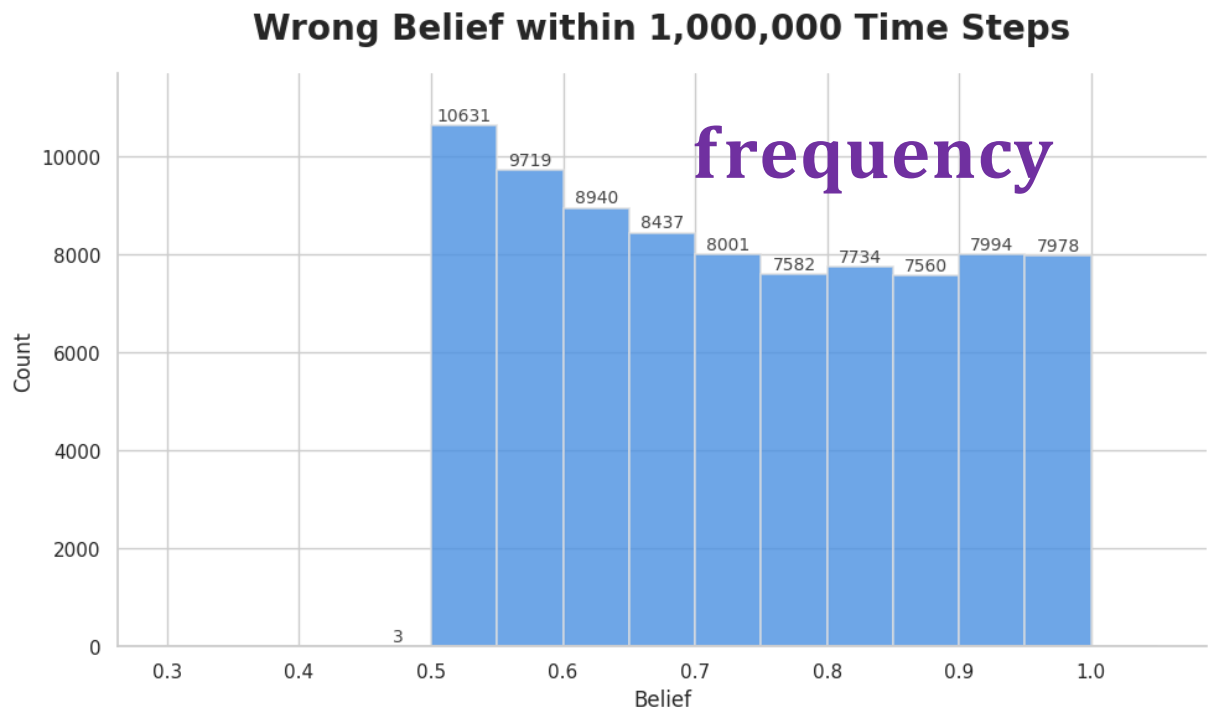
$$P(Y_0, Y_1, \dots, Y_k) = \sum_{i=0}^{2} \alpha_{k-1}\left(X^{(i)}\right)$$

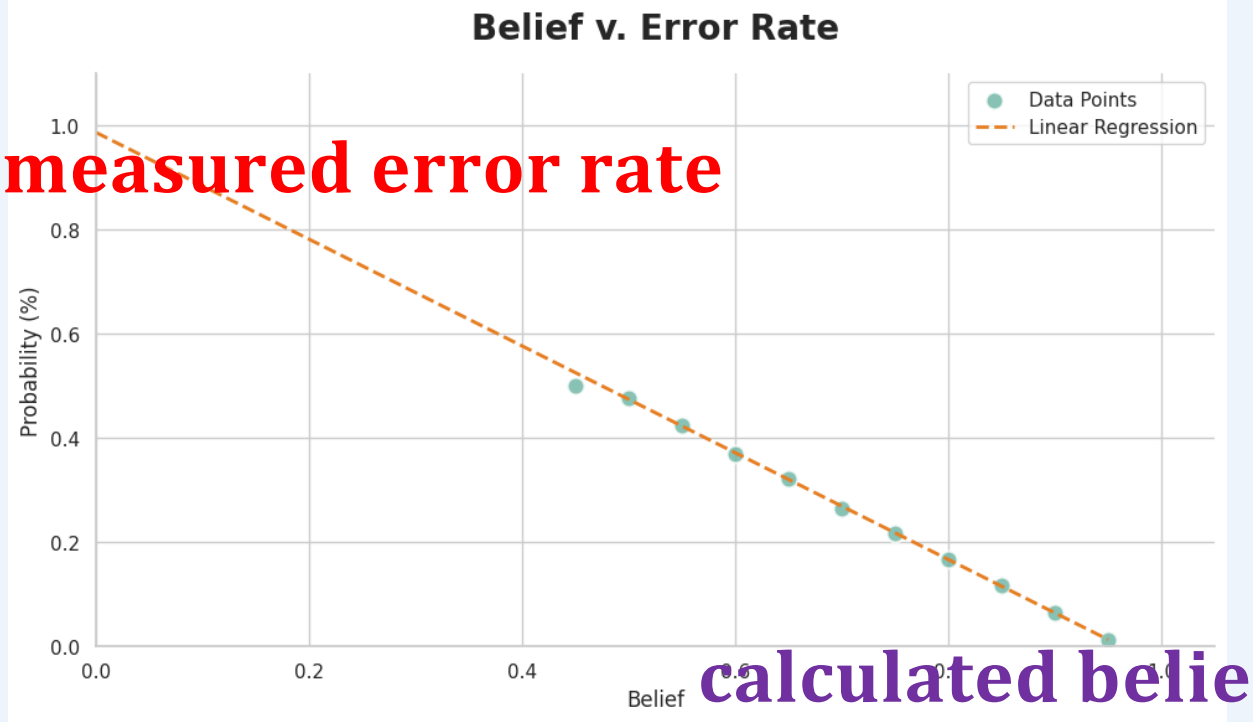
Forward Variable α_t / Belief $\hat{\pi}_t$

$$\alpha_0(X) = \overline{\pi} P(Y_k|X_k) = \overline{\pi}\Lambda_0 \rightarrow \hat{\pi}_k = \alpha_k / |\alpha_k|$$

$$\alpha_k(X) = \sum_{j=0}^{2} \alpha_{k-1}(X_{k-1}) P(X_k|X_{k-1}) P(Y_k|X_k) = \hat{\pi}_{k-1} M \Lambda_k$$

B. Probability – Robustness (1)

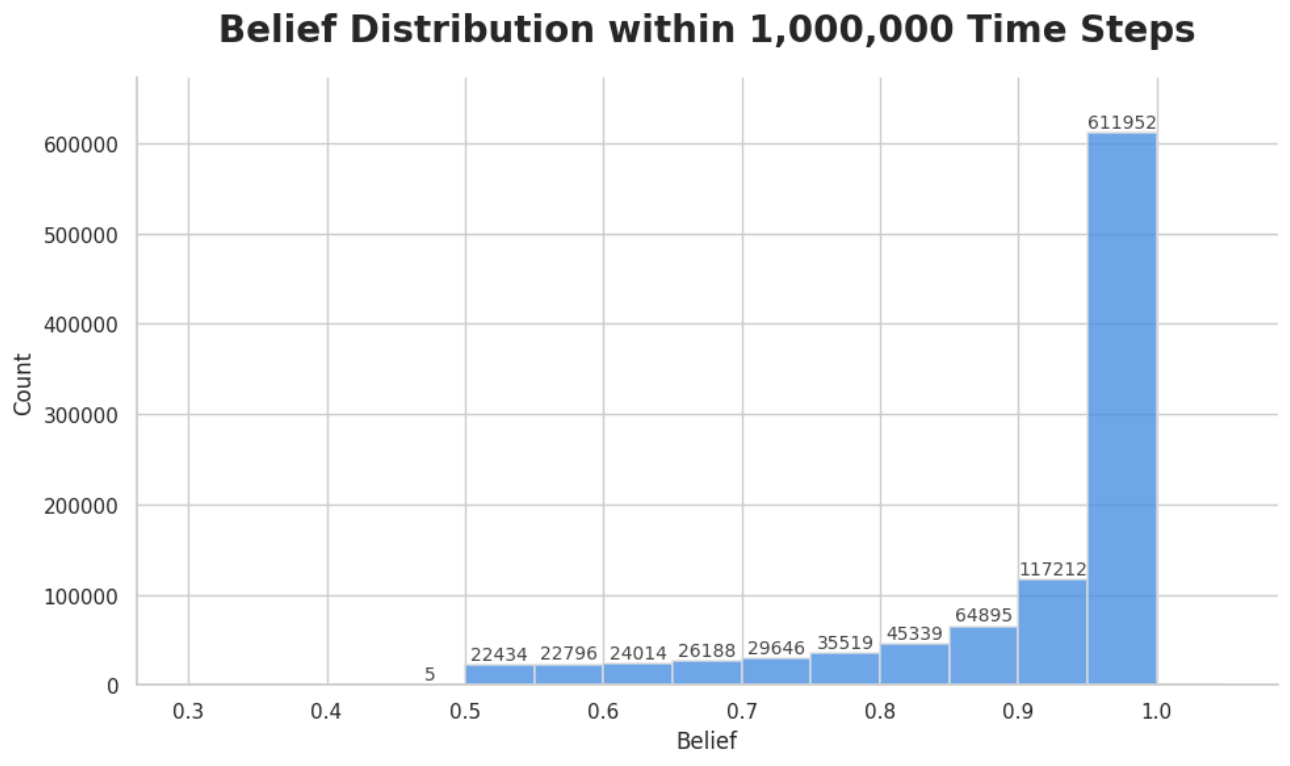


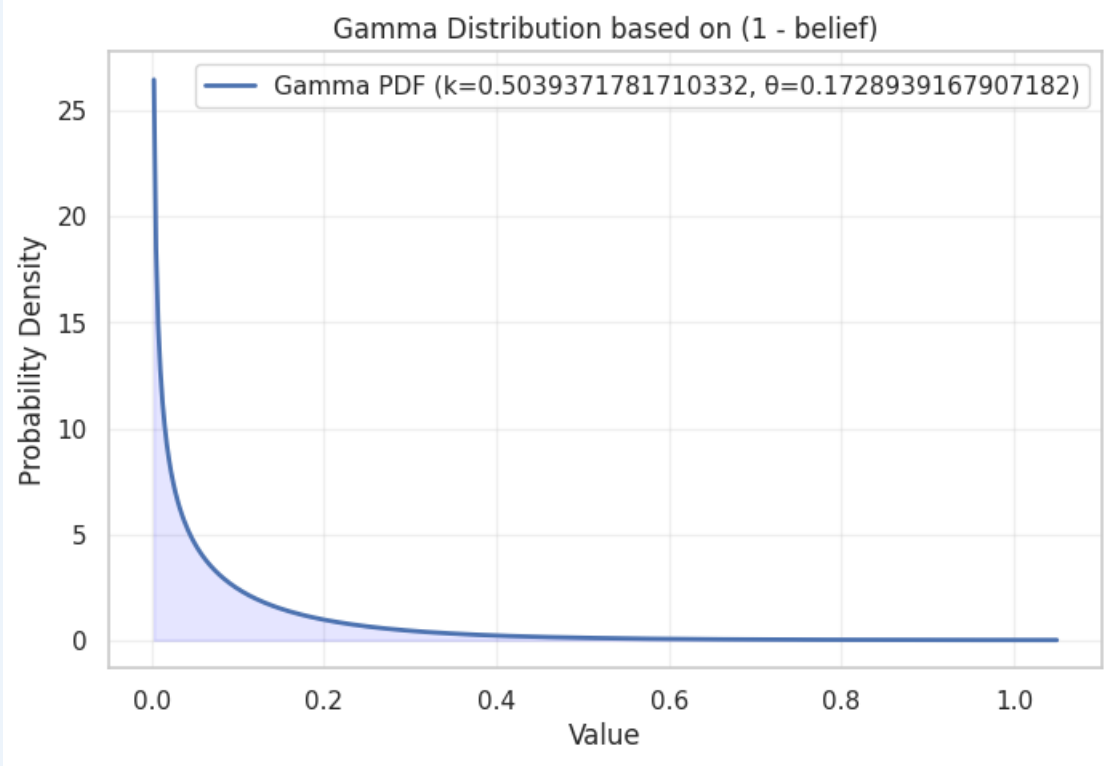


Cumulative numbers of error

$\hat{e} \approx 1 - \hat{\pi}$

B. Probability – Robustness (2)





Distribution of Belief:

→ *Model is with high precision* →

error belief

$$\hat{e} \approx 1 - \hat{\pi} \sim \Gamma(k = \text{shape} \approx 0.504, \theta = \text{scale} \approx 0.173)$$

8.43% 91.3%

C. Velocity Control Overview

Target Function:

- There exits a value function which can describe the target of maximizing completed tasks (in terms of displacement) and minimizing energy consumed simultaneously.
- The conclusion reveals there is always one optimal control under the provided belief at any state, regardless of the remaining power
- According to the previous conclusion, the key towards the power management must only be the task scheduling strategy.

C. Control – Suboptimal Diff

Value Function: $\boldsymbol{R_t} = \dfrac{v_t}{1 + 0.2(v_t - v^{(\hat{s}_t)})^2}$

s.t. $0 \le v_t \le 8$

velocity

consumption

$$\frac{d\boldsymbol{R_t}}{dv_t} = \frac{1 - 0.2\boldsymbol{v_t^2} + 0.2v^{(\hat{s}_t)^2}}{\cancel{\text{consumption}^2}} = 0 \rightarrow \boldsymbol{v_t^*} = \sqrt{5 + v^{(\hat{s}_t)^2}}$$

$$\boldsymbol{v^*} = \begin{cases} 8 & s = 0 \\ \sqrt{21} & s = 1 \\ \sqrt{5} & s = 2 \end{cases}$$

… when state is determined

C. Control – Optimal Solution

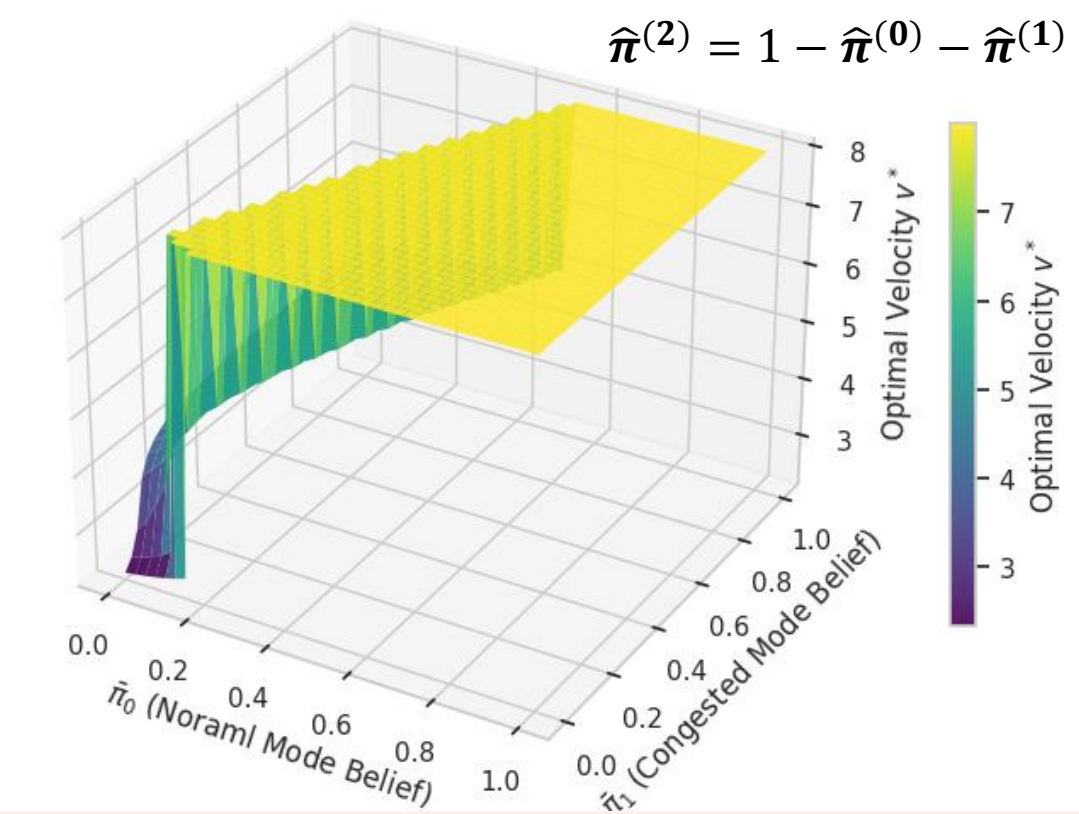
Value Function: $\overline{R}_t = \sum_{i=0}^{j} \frac{\hat{\pi}_t^{(i)} \cdot v_t}{1 + 0.2(v_t - v^{(i)})^2}$

s.t. $0 \leq v_t \leq 8$

Use Brent's method (minimize scalar tool from SciPy):

Find the minimal $-\overline{R}_t$

C. Control – Response Surface

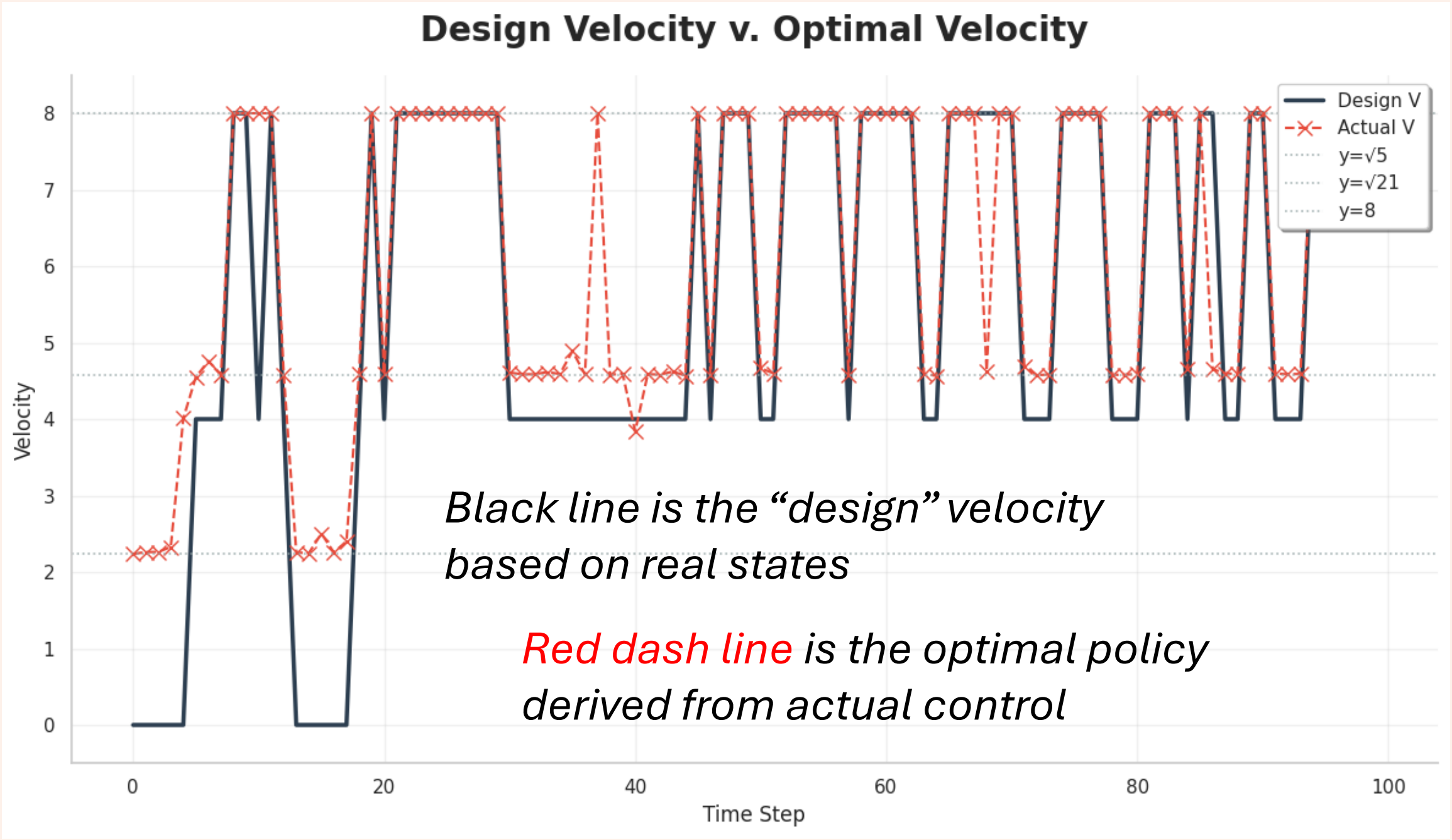


Avg for 720 steps in 100 runs

Plan	Optimal	Linear*
ΔX_{sum} +13%	556.3	492.7
ΔP_{sum} + 9%	982.8	898.5
Efficacy + 1%	0.57	0.55

$^* v_k = .1v'_k + .9v^{(\hat{s}_k)}$

$$\max_v \frac{\hat{\pi}^{(0)}v}{1+0.2(v-8)^2} + \frac{\hat{\pi}^{(1)}v}{1+0.2(v-4)^2} + \frac{\hat{\pi}^{(2)}v}{1+0.2v^2}$$

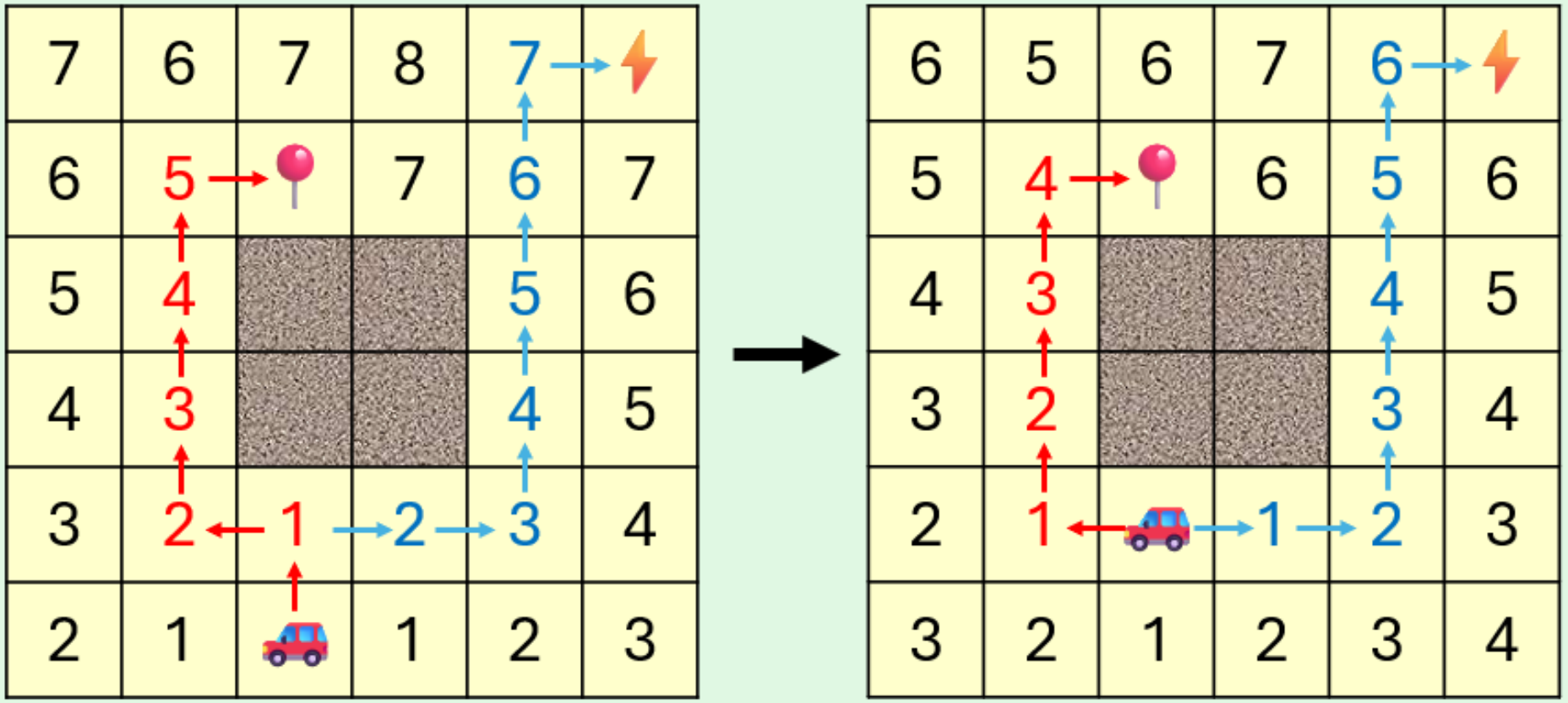

Design Velocity v. Optimal Velocity
Design V
Actual V
y=√5
y=√21
y=8
Velocity
Time Step
0
20
40
60
80
100
0
1
2
3
4
5
6
7
8
Black line is the "design" velocity based on real states
Red dash line is the optimal policy derived from actual control

D. Movement – Heuristic Strategy

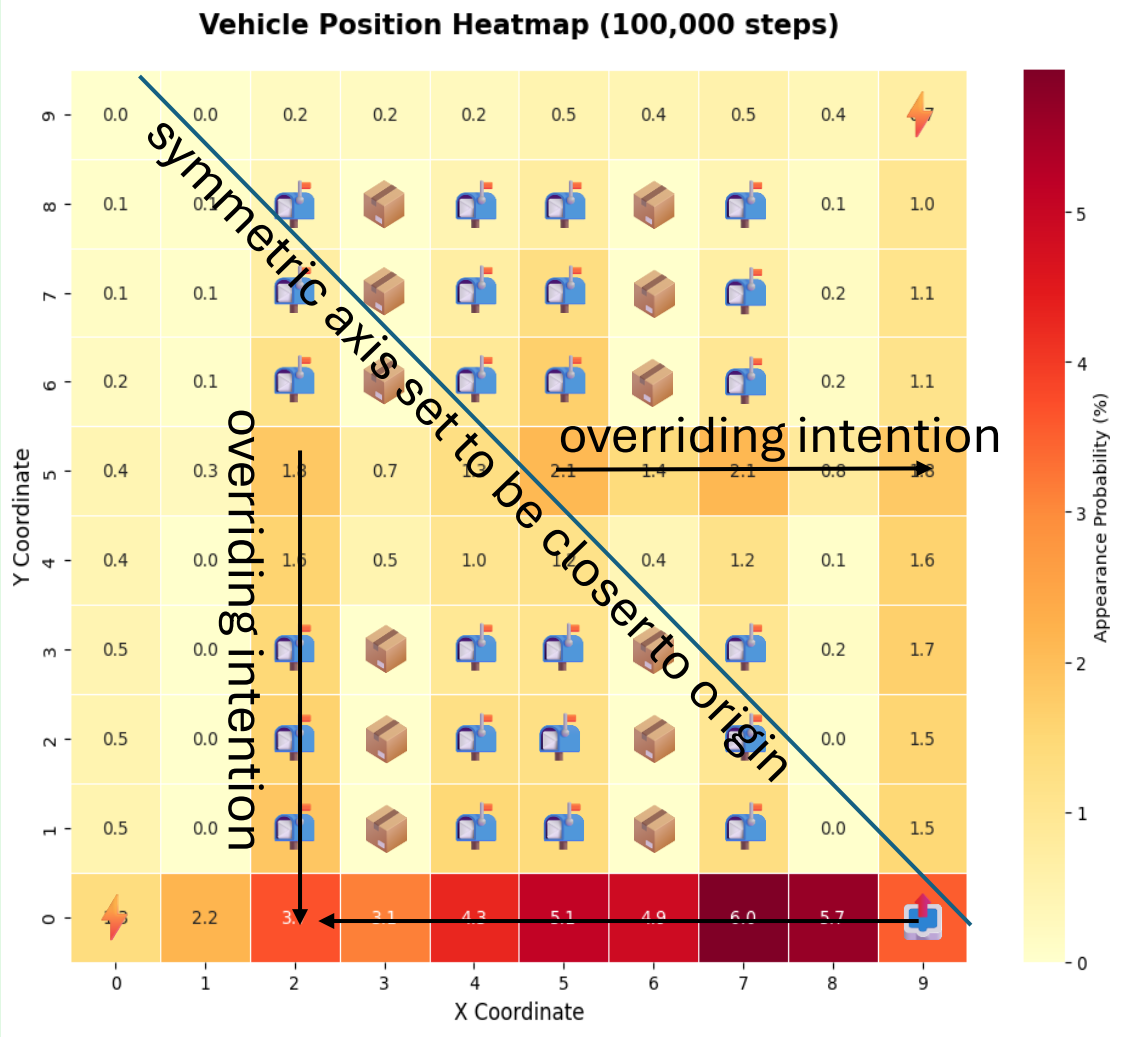
```
if nearest charger && next task in same direction:
    to task
elif power == low:
    to charger
elif power == medium && (belief == blocked ||
    distance(next task to charger − next charger) ≥ 5 &&
    distance(next task − next charger) ≥ 5):
    to charger
else:
    to task
```

Time Step 120~129
in Demonstration

D. Movement – Distance Judgement



BFS-assisted distance judgement


Vehicle Position Heatmap (100,000 steps)
symmetric axis set to be closer to origin
overriding intention
overriding intention
Y Coordinate
X Coordinate
Appearance Probability (%)


Average Vehicle Remaining Power Over Time in 1000 Runs
Power Level (%)
720 steps (1hr)
× 1000 runs
No fixed tendency → depends on the real states
Time Step v. Power Level in 1 Run
Power Level (%)
1 run
Time Steps

Thanks for your listening!

	Mean	STD
Task	58.79	4.51
v	6.17	1.95
ΔP	1.37	1.57
ΔP/Task	16.83	N/A

720 steps (1hr) × 1000 runs